

Technical Assessment – Machine Learning Engineer

Insuredmine

Instructions:

Make sure your final outputs are clean and well-commented.

Include a **README.md** explaining how to run your code.

Attempt all three sections.

Submission Format: ZIP file with code + output + README

Tools Allowed: Python, Jupyter/Colab, Internet, any public ML/NLP libraries

Section A – Premium Prediction :

Objective: Build a premium prediction model from structured insurance data.

♦ Problem Statement:

You are provided with a CSV file containing insurance premium records. The columns include:

- **Year, Month, Date**
- **Premium_Amount**

♦ Tasks:

1. Preprocess the data: Convert date components, encode features if necessary.
2. Engineer additional features: Example – sin/cos of month, etc.
3. Train a regression model (SVR, RandomForest, or XGBoost).

4. Predict premium as a percentage value.
5. Evaluate the model using R^2 and MAE.
6. Include a plot showing actual vs. predicted values.

Section B – NLP/OCR Text Extraction

Objective: Extract structured information from semi-structured OCR-like text.

♦ Problem Statement:

You are given a **.txt** file containing OCR output text of a customer information document (e.g., emails, names, addresses, phone numbers).

♦ Tasks:

1. Extract the following fields:
 - First Name
 - Last Name
 - Middle Name (optional)
 - Email
 - Phone Number
 - Date of Birth
 - Address
 - Marital Status
2. Return the result as a JSON object:
3. Handle multiple people in the same file.

4. Avoid duplicate entries and leave fields as "" or null if missing.

Section C – Quick Quiz

Answer briefly (2–4 lines).

1. What is the purpose of a vector database like FAISS in a RAG pipeline?
2. Which metric is best suited for regression model evaluation when outliers are present?
 - A) MAE
 - B) RMSE
 - C) R^2
 - D) Accuracy
3. What is the difference between Named Entity Recognition (NER) and Text Classification?
4. What are some advantages of using FastAPI over Flask in ML model deployment?
5. How would you secure a REST API that exposes a premium prediction endpoint?